

Supplement: Distance-based Analysis of Ordinal Data and Ordinal Time Series

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A Proofs

Note that empty sums $\sum_a^b$ with $b < a$ are assumed to take the value 0.

A.1 Some Useful Moment Formulae

It is well-known that the mean $E[I]$ of a count random variable I can be expressed as a sum about its CDF $(f_k)_{k \geq 0}$ by using the ‘‘Darth Vader rule’’. A more general result for expressing higher-order moments $E[I^r]$ (about the origin) in terms of the CDF was recently presented by Chakraborti et al. (2017). A slight simplification of their result is obtained if considering factorial moments $E[I_{(r)}]$ instead (where $x_{(r)} = x \cdots (x - r + 1)$ denotes the r^{th} falling factorial, $x_{(0)} := 1$):

$$E[I_{(r)}] = r \sum_{i \geq 0} i_{(r-1)} (1 - f_i) \quad \text{for } r \in \mathbb{N}. \quad (\text{A.1})$$

Proof: The proof of (A.1) is done in analogy to the derivation presented in Section 3 of Chakraborti et al. (2017). We start with the following identity:

$$i_{(r)} = \sum_{j=1}^i (j_{(r)} - (j-1)_{(r)}) = r \sum_{j=1}^i (j-1)_{(r-1)}.$$

Here, we first used a telescoping sum and the fact that $j_{(r)} = 0$ if $j < r$, and then the factorizations $j_{(r)} = j (j-1)_{(r-1)}$, $(j-1)_{(r)} = (j-1)_{(r-1)} (j-r)$. This identity is now used to show

$$\begin{aligned} E[I_{(r)}] &= \sum_{i \geq 1} i_{(r)} p_i = r \sum_{i \geq 1} \sum_{j=0}^{i-1} j_{(r-1)} p_i \\ &= r \sum_{j \geq 0} \sum_{i > j} j_{(r-1)} p_i = r \sum_{j \geq 0} j_{(r-1)} (1 - f_j), \end{aligned}$$

which completes the proof. #

In the sequel, we concentrate on count random variables I having the bounded range $\{0, \dots, m\}$ (rank count). Then (A.1) becomes

$$E[I_{(r)}] = m_{(r)} - r \sum_{i=0}^{m-1} i_{(r-1)} f_i \quad \text{for } r \in \mathbb{N}. \quad (\text{A.2})$$

Proof: Since $1 - f_m = 0$, (A.1) implies

$$E[I_{(r)}] = r \sum_{i=0}^{m-1} i_{(r-1)} (1 - f_i) \quad \text{for } r \in \mathbb{N}.$$

Using the well-known identity $\sum_{k=0}^n k_{(r)} = \frac{(n+1)_{(r+1)}}{r+1}$, it follows that

$$\sum_{i=0}^{m-1} i_{(r-1)} = \frac{m_{(r)}}{r},$$

which completes the proof of (A.2). #

Using (A.2), the variance of I can be expressed as

$$V[I] = \sum_{i=0}^{m-1} f_i (1 - \sum_{j=0}^{m-1} f_j) + 2 \sum_{i=1}^{m-1} \sum_{j=0}^{i-1} f_j. \quad (\text{A.3})$$

Proof: Using (A.2), we write

$$\begin{aligned} V[I] &= E[I_{(2)}] + E[I] - E[I]^2 \\ &= m_{(2)} - 2 \sum_{i=0}^{m-1} i f_i + (m - \sum_{i=0}^{m-1} f_i) - (m - \sum_{i=0}^{m-1} f_i)^2 \\ &= -2 \sum_{i=0}^{m-1} i f_i + (2m - 1) \sum_{i=0}^{m-1} f_i - \sum_{i,j=0}^{m-1} f_i f_j. \end{aligned}$$

Then we compute

$$\begin{aligned} \sum_{i=0}^{m-1} i f_i &= \sum_{i=1}^{m-1} \sum_{j=1}^i f_i = \sum_{j=1}^{m-1} \sum_{i=j}^{m-1} f_i \\ &= \sum_{j=1}^{m-1} (\sum_{i=0}^{m-1} f_i - \sum_{i=0}^{j-1} f_i) \\ &= (m-1) \sum_{i=0}^{m-1} f_i - \sum_{j=1}^{m-1} \sum_{i=0}^{j-1} f_i. \end{aligned}$$

Inserting into the formula for $V[I]$ completes the proof. #

A.2 Derivations of Section 3.2

According to (5), the DIVC is defined as $\text{disp}_d(X) := E[d(X_1, X_2)]$, where X_1, X_2 are two independent copies of X , having the range $\mathcal{S} = \{s_0, \dots, s_m\}$. As long as X does not follow a one-point distribution, there exists at least one truly positive product $p_i \cdot p_j$ with $i \neq j$. So if the distance measure d is positive definite (see Definition 2.1), then also

$$\text{disp}_d = 2 \sum_{i < j} d(s_i, s_j) p_i p_j$$

is truly positive. So for a positive definite distance measure d , disp_d takes its minimal value 0 exactly for a one-point distribution.

Next, we consider the question when disp_d becomes maximal.

A.2.1 Lemma *Consider disp_d as a function of $p_0, \dots, p_{m-1}$ (with $p_m = 1 - p_0 - \dots - p_{m-1}$) on the m -part open simplex. Then the first-order partial derivatives are given by*

$$\frac{\partial}{\partial p_k} \text{disp}_d(X) = 2 \left(E[d(X, s_k)] - E[d(X, s_m)] \right) \quad \text{for } k = 0, \dots, m-1.$$

Note that we have $\frac{\partial}{\partial p_0} \text{disp}_d = 0$ iff the skewness measure skew_d from (12) equals 0.

Proof: We rewrite disp_d as

$$\text{disp}_d = 2 \sum_{i=0}^{m-2} \sum_{j=i+1}^{m-1} d(s_i, s_j) p_i p_j + 2 \sum_{i=0}^{m-1} d(s_i, s_m) p_i (1 - p_0 - \dots - p_{m-1}),$$

i.e., as a quadratic polynomial in $p_0, \dots, p_{m-1}$. Then we compute (by considering the summands with either $i = k$ or $j = k$)

$$\begin{aligned} \frac{\partial}{\partial p_k} \text{disp}_d &= 2 \sum_{j=k+1}^{m-1} d(s_k, s_j) p_j + 2 d(s_k, s_m) (1 - 2p_k - \sum_{r=0, r \neq k}^{m-1} p_r) \\ &\quad + 2 \sum_{i=0}^{k-1} d(s_i, s_k) p_i + 2 \sum_{i=0, i \neq k}^{m-1} d(s_i, s_m) p_i (-1). \end{aligned}$$

Using that $1 - 2p_k - \sum_{r=0, r \neq k}^{m-1} p_r = p_m - p_k$, it follows that

$$\begin{aligned} \frac{\partial}{\partial p_k} \text{disp}_d &= 2 \sum_{i=0, i \neq k}^m d(s_i, s_k) p_i - 2 \sum_{i=0}^{m-1} d(s_i, s_m) p_i \\ &= 2 \sum_{i=0}^m d(s_i, s_k) p_i - 2 \sum_{i=0}^m d(s_i, s_m) p_i, \end{aligned}$$

because $d(s_k, s_k) = 0 = d(s_m, s_m)$. This completes the proof. #

Lemma A.2.1 can now be used to derive the maximum of disp_d . If applied to the Hamming distance (1), the partial derivatives $\frac{\partial}{\partial p_k} \text{disp}_{d_H}$ become 0 iff all expectations

$$E[d_H(X, s_k)] = E[1 - \delta_{X, s_k}] = 1 - p_k \quad \text{for } k = 0, \dots, m$$

coincide, i.e., if X follows the uniform distribution on $\mathcal{S}$ (where $p_k = \frac{1}{m+1}$). This well-known result (see the discussion of Gini index and IQV in Section 3.2) implies that the maximal value of disp_{d_H} is given by $\frac{m}{m+1}$ such that IQV is normalized with range $[0; 1]$.

We now consider a family of distance measures being more relevant in the ordinal case.

A.2.2 Theorem *Let X be an ordinal random variable, and let d be additive according to property (O3) in Definition 2.2.*

Then disp_d is maximized in the extreme two-point distribution (where $p_0 = p_m = \frac{1}{2}$) with maximal value $\frac{1}{2} d(s_0, s_m)$.

Note that Theorem A.2.2 covers the important special case of the ordinal block distance (2), where $d_1 = \dots = d_m = 1$.

Proof: Lemma A.2.1 leads to a system of linear equations in $p_0, \dots, p_{m-1}$:

$$\begin{aligned}
& \frac{\partial}{\partial p_k} \text{disp}_d(X) = 0 \\
\Leftrightarrow & E[d(X, s_k)] = E[d(X, s_m)] \\
\Leftrightarrow & \sum_{i=0}^{m-1} d(s_i, s_k) p_i + d(s_k, s_m) (1 - \sum_{j=0}^{m-1} p_j) = \sum_{i=0}^{m-1} d(s_i, s_m) p_i \\
\Leftrightarrow & \sum_{i=0}^{m-1} (d(s_i, s_m) + d(s_k, s_m) - d(s_i, s_k)) p_i = d(s_k, s_m) \\
\Leftrightarrow & \sum_{i=0}^{m-1} 2(d_{\max\{i,k\}+1} + \dots + d_m) p_i = d_{k+1} + \dots + d_m
\end{aligned}$$

for $k = 0, \dots, m-1$. So the resulting coefficient matrix $\mathbf{A}$ has the entries $a_{r,s} = 2(d_{\max\{r,s\}} + \dots + d_m)$ for $r, s = 1, \dots, m$. The corresponding inverse matrix $\mathbf{A}^{-1}$ is a symmetric tridiagonal matrix with main diagonal elements $\frac{1}{2d_1}, \frac{1}{2d_1} + \frac{1}{2d_2}, \dots, \frac{1}{2d_{m-1}} + \frac{1}{2d_m}$, and first diagonals' elements $-\frac{1}{2d_1}, \dots, -\frac{1}{2d_{m-1}}$, because

$$\begin{aligned}
(\mathbf{A} \mathbf{A}^{-1})_{r,1} &= \frac{1}{2d_1} (a_{r,1} - a_{r,2}) = \frac{d_1 \delta_{r,1}}{d_1} = \delta_{r,1}, \\
(\mathbf{A} \mathbf{A}^{-1})_{r,s} &= \frac{1}{2d_{s-1}} (a_{r,s} - a_{r,s-1}) + \frac{1}{2d_s} (a_{r,s} - a_{r,s+1}) \\
&= -\frac{\mathbb{1}_{\{r < s\}} d_{s-1}}{d_{s-1}} + \frac{\mathbb{1}_{\{r \leq s\}} d_s}{d_s} = \delta_{r,s} \quad \text{for } s = 2, \dots, m-1, \\
(\mathbf{A} \mathbf{A}^{-1})_{r,m} &= \frac{1}{2d_{m-1}} (a_{r,m} - a_{r,m-1}) + \frac{1}{2d_m} a_{r,m} \\
&= -\frac{\mathbb{1}_{\{r < m\}} d_{m-1}}{d_{m-1}} + \frac{d_m}{d_m} = \delta_{r,m}.
\end{aligned}$$

Here, we used that

$$\frac{1}{2} (a_{r,s} - a_{r,s+1}) = (d_{\max\{r,s\}} + \dots + d_m) - (d_{\max\{r,s+1\}} + \dots + d_m) = \mathbb{1}_{\{r \leq s\}} d_s.$$

Finally, the above system of linear equations is solved by multiplying $\mathbf{A}^{-1}$ to its right-hand

side, leading to

$$\begin{aligned}
& \begin{pmatrix} \frac{1}{2d_1} & \frac{-1}{2d_1} & & 0 \\ \frac{-1}{2d_1} & \frac{1}{2d_1} + \frac{1}{2d_2} & \ddots & \\ & \ddots & \ddots & \frac{-1}{2d_{m-1}} \\ 0 & & \frac{-1}{2d_{m-1}} & \frac{1}{2d_{m-1}} + \frac{1}{2d_m} \end{pmatrix} \begin{pmatrix} d_1 + d_2 + \dots + d_m \\ d_2 + \dots + d_m \\ \vdots \\ d_m \end{pmatrix} \\
&= \begin{pmatrix} \frac{d_1}{2d_1} \\ \vdots \\ \frac{-d_{r-1}}{2d_{r-1}} + \frac{d_r}{2d_r} \\ \vdots \\ \frac{-d_{m-1}}{2d_{m-1}} + \frac{d_m}{2d_m} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 0 \\ \vdots \\ 0 \end{pmatrix},
\end{aligned}$$

which is the extreme two-point distribution $\mathbf{p}_{\text{two}}$. In this case, $\text{disp}_d = 2d(s_0, s_m) \frac{1}{2^2} = \frac{1}{2} d(s_0, s_m)$. This completes the proof. $\#$

If considering the ordinal squared Euclidean distance (3) instead, Lemma A.2.1 implies that there does not exist an extremal point with zero derivatives: Since

$$\frac{\partial}{\partial p_k} \text{disp}_{d_{o,2}} = 2 \left(E[(I - k)^2] - E[(I - m)^2] \right) = 2(m - k) \left(2E[I] - m - k \right)$$

for $k = 0, \dots, m - 1$, $\frac{\partial}{\partial p_0} \text{disp}_{d_{o,2}} = 0$ implies that $E[I] = \frac{m}{2}$ has to hold. This, in turn, implies that the remaining derivatives are truly negative:

$$\frac{\partial}{\partial p_k} \text{disp}_{d_{o,2}} = -2k(m - k) < 0 \quad \text{for } k = 1, \dots, m - 1.$$

But since $\text{disp}_{d_{o,2}}$ decreases in $p_1, \dots, p_{m-1}$, $\text{disp}_{d_{o,2}}$ is maximized only if $p_1 = \dots = p_{m-1} = 0$. Together with $E[I] = \frac{m}{2}$, this implies that also $\text{disp}_{d_{o,2}}$ from (8) is maximized for the extreme two-point distribution $\mathbf{p}_{\text{two}}$, with the maximal value of $\text{disp}_{d_{o,2}} = 2(E[I^2] - E[I]^2)$ being $2\left(\frac{m^2}{2} - \frac{m^2}{4}\right) = \frac{m^2}{2}$.

A.3 Derivations of Section 3.3

The upper bound $d(s_0, s_m)$ for asym_d is reached for the extreme one-point distributions $p_0 = 1$ or $p_m = 1$, but it is sometimes also reached for further distributions.

A.3.1 Example If using the Hamming distance, then we have $E[d_H(X, X^r)] = 1 - \sum_{i=0}^m p_i p_{m-i}$, which equals $1 - s_2(\mathbf{p}) = \text{disp}_{d_H}$ for a symmetric distribution. If, in contrast, $p_i = 0$ for $i \geq \lceil \frac{m+1}{2} \rceil$, then we obtain the value $E[d_H(X, X^r)] = 1$; the same happens if $p_i = 0$ for $i \leq \lfloor \frac{m-1}{2} \rfloor$. So any non-central one-point distribution leads to $\text{asym}_{d_H} = 1$.

The following lemma prepares our analyses of the asymmetry measure asym_d from (9).

A.3.2 Lemma *Let X, X' and Y, Y' be independent ordinal random variables defined on the same ordinal range $\mathcal{S} = \{s_0, \dots, s_m\}$ with distance measure d and corresponding distance matrix $\mathbf{D} = (d(s_i, s_j))_{i,j=0,\dots,m}$. Let the PMF of X, X' be given by $\mathbf{p}$, the one of Y, Y' by $\boldsymbol{\pi}$.*

Then it holds that

$$E[d(X, X')] + E[d(Y, Y')] - 2 E[d(X, Y)] = (\mathbf{p} - \boldsymbol{\pi})^\top \mathbf{D} (\mathbf{p} - \boldsymbol{\pi}).$$

Note that $E[d(X, X')]$ and $E[d(Y, Y')]$ are the DIVCs of X and Y , respectively. Also note that the (real) eigenvalues of $\mathbf{D}$ sum up to 0 because of the zero diagonal (hollow symmetric matrix), i. e., $\mathbf{D}$ is generally an indefinite matrix.

Proof: Since a distance measure is symmetric (D2), we have $E[d(X, Y)] = E[d(Y, X)]$, and we obtain

$$\begin{aligned} & E[d(X, X')] + E[d(Y, Y')] - E[d(X, Y)] - E[d(Y, X)] \\ &= \sum_{i,j=0}^m \left(d(s_i, s_j) p_i p_j + d(s_i, s_j) \pi_i \pi_j - d(s_i, s_j) p_i \pi_j - d(s_i, s_j) \pi_i p_j \right) \\ &= \sum_{i,j=0}^m d(s_i, s_j) (p_i - \pi_i)(p_j - \pi_j). \end{aligned}$$

This completes the proof of Lemma A.3.2. #

We now apply Lemma A.3.2 to the random variables X, X^r . As argued in the beginning of Section 3.3, if d is centrosymmetric (O4), then the DIVCs of X and X^r are equal to each other. So it follows that

$$\text{asym}_d = -\frac{1}{2} \sum_{i,j=0}^m d(s_i, s_j) (p_i - p_{m-i})(p_j - p_{m-j}),$$

which equals 0 for a symmetric distribution. Let $\mathbf{J} = (\delta_{i,m-j})_{i,j=0,\dots,m}$ denote the $(m+1)$ -dimensional exchange matrix, which satisfies $\mathbf{J}^2 = \mathbf{I}$. Then $\mathbf{J}\mathbf{p}$ is the reflection of $\mathbf{p}$, and the centrosymmetry of $\mathbf{D}$ implies that $\mathbf{D}\mathbf{J} = \mathbf{J}\mathbf{D}$. So we can further write

$$\text{asym}_d = -\frac{1}{2} (\mathbf{p} - \mathbf{J}\mathbf{p})^\top \mathbf{D} (\mathbf{p} - \mathbf{J}\mathbf{p}) = \mathbf{p}^\top (\mathbf{J} - \mathbf{I}) \mathbf{D} \mathbf{p}$$

as a quadratic form in $\mathbf{p}$ (whereas $\text{disp}_d = \mathbf{p}^\top \mathbf{D} \mathbf{p}$). Therefore, if $(\mathbf{J} - \mathbf{I})\mathbf{D}$ is positive semi-definite, then the minimal value of asym_d is given by 0.

Because of being both symmetric and centrosymmetric, the eigenvalues and definiteness of $(\mathbf{J} - \mathbf{I})\mathbf{D}$ can be checked by utilizing Lemma 3 in Cantoni & Butler (1976). Consider the blocks $\mathbf{A} = (d_{m-i,j} - d_{i,j})_{i,j=0,\dots,\lfloor \frac{m-1}{2} \rfloor}$ and $\mathbf{C} = (d_{m-i,j} - d_{i,j})_{i=\lceil \frac{m+1}{2} \rceil, \dots, m, j=0,\dots,\lfloor \frac{m-1}{2} \rfloor}$ taken from $\mathbf{D}$, and define $\tilde{\mathbf{J}} = (\delta_{i,m-j})_{i,j=0,\dots,\lfloor \frac{m-1}{2} \rfloor}$. Then $\mathbf{A} = -\tilde{\mathbf{J}}\mathbf{C}$, so using Cantoni & Butler (1976), the non-zero eigenvalues of $(\mathbf{J} - \mathbf{I})\mathbf{D}$ are also eigenvalues of

$$\mathbf{A} - \tilde{\mathbf{J}}\mathbf{C} = 2\mathbf{A} = 2(d_{m-i,j} - d_{i,j})_{i,j=0,\dots,\lfloor \frac{m-1}{2} \rfloor}.$$

A.3.3 Example If d equals the Hamming distance, then $\mathbf{D}$ has the entries $d_{i,j} = 1 - \delta_{i,j}$, so $2\mathbf{A} = 2(\delta_{i,j})_{i,j=0,\dots,\lfloor \frac{m-1}{2} \rfloor}$. Thus, all eigenvalues are equal to 2.

We already know from (10) that $\text{asym}_{d_{0,2}} \geq 0$ also holds for the squared Euclidean distance (3). This is cross-checked as follows: Since

$$2a_{i,j} = 2((m-2i)(m-2j)),$$

the last column of $\mathbf{A}$ (the one for $j = \lfloor \frac{m-1}{2} \rfloor$) has the entries $m-2i$ such that the remaining columns are multiples of the last column. Consequently, all but one eigenvalue are equal to 0. The unique non-zero eigenvalue of $2\mathbf{A}$ corresponds to the eigenvector with entries $m-2j$, and it equals $2 \sum_{j=0}^{\lfloor \frac{m-1}{2} \rfloor} (m-2j)^2 > 0$.

A simple counterexample, where asym_d might become negative, is obtained by using fourth powers for the distances. Let $d_{0,4}(s_i, s_j) = (i-j)^4$ and let $m = 3$, then it is easily computed that

$$\mathbf{D} = \begin{pmatrix} 0 & 1 & 16 & 81 \\ 1 & 0 & 1 & 16 \\ 16 & 1 & 0 & 1 \\ 81 & 16 & 1 & 0 \end{pmatrix}, \quad (\mathbf{J} - \mathbf{I})\mathbf{D} = \begin{pmatrix} 81 & 15 & -15 & -81 \\ 15 & 1 & -1 & -15 \\ -15 & -1 & 1 & 15 \\ -81 & -15 & 15 & 81 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 81 & 15 \\ 15 & 1 \end{pmatrix}.$$

The eigenvalues of $2\mathbf{A}$ are $2(41 \pm 5\sqrt{73})$, so one of them being negative. In accordance with $(\mathbf{J} - \mathbf{I})\mathbf{D}$ being indefinite, counterexamples are easily constructed: let $\mathbf{p} = (0, 0.8, 0, 0.2)^\top$, then

$$\text{asym}_d = \mathbf{p}^\top \mathbf{D} \mathbf{J} \mathbf{p} - \mathbf{p}^\top \mathbf{D} \mathbf{p} = 4.20 - 5.12 < 0.$$

The expressions given for $\text{asym}_{d_{o,1}}$ in formula (11) are derived as follows:

$$\begin{aligned}
E[d(X, X^r)] &= \sum_{i=0}^{m-1} \sum_{j=i+1}^m (j-i) (p_i p_{m-j} + p_j p_{m-i}) \\
&= \sum_{i=0}^{m-1} \sum_{j=i+1}^m \sum_{k=i+1}^j (p_i p_{m-j} + p_j p_{m-i}) \\
&= \sum_{i=0}^{m-1} \sum_{k=i+1}^m \sum_{j=k}^m (p_i p_{m-j} + p_j p_{m-i}) \\
&= \sum_{i=0}^{m-1} \sum_{k=i+1}^m (p_i f_{m-k} + p_{m-i} (1 - f_{k-1})) \\
&= \sum_{j=0}^{m-1} \sum_{i=0}^j (p_i f_{m-j-1} + p_{m-i} (1 - f_j)) \\
&= \sum_{j=0}^{m-1} (f_j f_{m-j-1} + (1 - f_{m-j-1}) (1 - f_j)).
\end{aligned}$$

Using that $\text{disp}_{d_{o,1}} = \sum_{j=0}^{m-1} f_j (1 - f_j) + \sum_{j=0}^{m-1} f_{m-j-1} (1 - f_{m-j-1})$, see (7), we obtain that

$$\text{asym}_{d_{o,1}}(X) = E[d(X, X^r)] - \text{disp}_{d_{o,1}}(X) = \sum_{j=0}^{m-1} (1 - f_j - f_{m-j-1})^2.$$

Furthermore, it also follows that

$$\begin{aligned}
\text{asym}_{d_{o,1}} + 2 \text{disp}_{d_{o,1}} &= \sum_{j=0}^{m-1} (1 + 2 f_j f_{m-j-1} - f_j^2 - f_{m-j-1}^2) \\
&= \sum_{j=0}^{m-1} (1 - (f_j - f_{m-j-1})^2) \in [0; m].
\end{aligned}$$

So $\frac{1}{m} \text{asym}_{d_{o,1}} \leq 1 - \frac{2}{m} \text{disp}_{d_{o,1}} = 1 - \text{IOV}$. Analogously,

$$\text{asym}_{d_{o,2}} + 2 \text{disp}_{d_{o,2}} = E[(2I - m)^2] \in [0; m^2],$$

so the normalized measures satisfy $\frac{1}{m^2} \text{asym}_{d_{o,2}} \leq 1 - \frac{2}{m^2} \text{disp}_{d_{o,2}}$.

The measure skew_d from (12) can be rewritten as

$$\begin{aligned}
\text{skew}_d &= \sum_{i=0}^{m-1} (d(s_i, s_m) - d(s_i, s_0)) p_i - d(s_m, s_0) (1 - p_0 - \dots - p_{m-1}) \\
&= \sum_{i=0}^{m-1} (d(s_i, s_m) + d(s_m, s_0) - d(s_i, s_0)) p_i - d(s_m, s_0),
\end{aligned}$$

also see the proof of Lemma A.2.1 (setting $k = 0$). The p_i 's coefficients are non-negative if the triangular inequality (D3) is satisfied or if the maximization property (O1) holds (then $d(s_m, s_0) - d(s_i, s_0) \geq 0$). Then skew_d becomes minimal if $p_0 = \dots = p_{m-1} = 0$ (extreme right one-point distribution), taking the value $-d(s_0, s_m)$. Analogously, one argues that skew_d takes the maximal value $+d(s_0, s_m)$ for $p_0 = 1$.

Furthermore, we can also write

$$\text{skew}_d = \sum_{i=0}^m d(s_{m-i}, s_m) p_{m-i} - \sum_{i=0}^m d(s_i, s_0) p_i.$$

If d is centrosymmetric (O4), then this expression becomes

$$\text{skew}_d = \sum_{i=0}^m d(s_i, s_0) p_{m-i} - \sum_{i=0}^m d(s_i, s_0) p_i = \sum_{i=0}^m d(s_i, s_0) (p_{m-i} - p_i).$$

So if the distribution is symmetric, this becomes 0 as $p_{m-i} - p_i = 0$ for all i .

Equation (13) is shown by computing

$$\begin{aligned} \frac{1}{m} \text{skew}_{d_{o,1}} &= \frac{1}{m} E[d_{o,1}(X, s_m)] - \frac{1}{m} E[d_{o,1}(X, s_0)] \\ &= \frac{1}{m} E[m - I] - \frac{1}{m} E[I - 0] = 1 - \frac{2}{m} E[I]. \end{aligned}$$

Note that $E[I] = m - \sum_{k=0}^{m-1} f_k$ because of (A.2), so

$$\frac{1}{m} \text{skew}_{d_{o,1}} = 1 - \frac{2}{m} E[I] = \frac{2}{m} \sum_{i=0}^{m-1} f_i - 1.$$

The measure $S^K = S^{b_1}$ proposed by Klein & Doll (2018), in turn, is given by

$$\frac{1}{m} \sum_{i=0}^{m-1} (f_i + f_{m-1-i} - 1) = \frac{2}{m} \sum_{i=0}^{m-1} f_i - 1.$$

A.4 Derivations of Section 6

To prove equation (19), we compute

$$\begin{aligned}
E[|I_X - I_Y|] &= \sum_{l=0}^{m-1} \sum_{k=l+1}^m (k-l) \left(P(I_X = k, I_Y = l) + P(I_X = l, I_Y = k) \right) \\
&= \sum_{l=0}^{m-1} \sum_{k=l+1}^m \sum_{j=l+1}^k \left(P(I_X = k, I_Y = l) + P(I_X = l, I_Y = k) \right) \\
&= \sum_{l=0}^{m-1} \sum_{j=l+1}^m \sum_{k=j}^m \left(P(I_X = k, I_Y = l) + P(I_X = l, I_Y = k) \right) \\
&= \sum_{l=0}^{m-1} \sum_{j=l+1}^m \left(P(I_X \geq j, I_Y = l) + P(I_X = l, I_Y \geq j) \right) \\
&= \sum_{l=0}^{m-1} \sum_{k=l}^{m-1} \left(P(I_X > k, I_Y = l) + P(I_X = l, I_Y > k) \right) \\
&= \sum_{k=0}^{m-1} \sum_{l=0}^k \left(P(I_X > k, I_Y = l) + P(I_X = l, I_Y > k) \right) \\
&= \sum_{k=0}^{m-1} \left(P(I_X > k, I_Y \leq k) + P(I_X \leq k, I_Y > k) \right).
\end{aligned}$$

Since $P(I_X > k, I_Y \leq k) = P(I_Y \leq k) - P(I_X \leq k, I_Y \leq k)$ and $P(I_X \leq k, I_Y > k) = P(I_X \leq k) - P(I_X \leq k, I_Y \leq k)$, we obtain that

$$E[|I_X - I_Y|] = \sum_{k=0}^{m-1} \left(P(I_X \leq k) + P(I_Y \leq k) - 2P(I_X \leq k, I_Y \leq k) \right).$$

A.5 A Central Limit Theorem

We consider the asymptotic distribution of such ordinal measures which rely on the (bivariate) CDF of the ordinal process or ordinal i.i.d. sample. Estimation is done by replacing the true (bivariate) CDF,

$$f_i = P(I_t \leq i), \quad f_{ij}^{(h)} = P(I_t \leq i, I_{t-h} \leq j),$$

by the corresponding sample counterpart. The asymptotic distribution of these sample measures is then derived with an analogous approach as in Kiesl (2003); Weiß (2013). For

this purpose, we consider the indicators $\mathbb{1}_{\{I_t \leq i\}}$ for $i \in \{0, \dots, m-1\}$, and we define the $2m$ -dimensional vectors

$$\begin{aligned} \mathbf{Z}_t^{(h)} &= (Z_{t;0}, \dots, Z_{t;m-1}, Z_{t;m+0}^{(h)}, \dots, Z_{t;m+(m-1)}^{(h)})^\top \quad \text{for lags } h \geq 1, \\ \text{where } Z_{t;i} &= \mathbb{1}_{\{I_t \leq i\}}, & \text{for } i = 0, \dots, m-1, \\ Z_{t;m+i}^{(h)} &= \mathbb{1}_{\{I_t \leq i\}} \mathbb{1}_{\{I_{t-h} \leq i\}}, \end{aligned} \tag{A.4}$$

and

$$\begin{aligned} \mathbf{Y}_t^{(h)} &= (Y_{t;0}, \dots, Y_{t;m-1}, Y_{t;m+0}^{(h)}, \dots, Y_{t;m+(m-1)}^{(h)})^\top \quad \text{for lags } h \geq 1, \\ \text{where } Y_{t;i} &= Z_{t;i} - f_i, & \text{for } i = 0, \dots, m-1. \\ Y_{t;m+i}^{(h)} &= Z_{t;m+i}^{(h)} - f_{ii}^{(h)}, \end{aligned} \tag{A.5}$$

Note that $\mathbb{1}_{\{I_t \leq i\}} \mathbb{1}_{\{I_{t-h} \leq i\}} = \mathbb{1}_{\{I_t \leq i, I_{t-h} \leq i\}}$ holds. By construction, we have $E[\mathbf{Y}_t^{(h)}] = \mathbf{0}$. Furthermore, the empirical (bivariate) CDF corresponding to the time series $X_1, \dots, X_n$ is given by

$$\hat{f}_i = \frac{1}{n} \sum_{t=1}^n \mathbb{1}_{\{I_t \leq i\}}, \quad \hat{f}_{ij}^{(h)} = \frac{1}{n-h} \sum_{t=h+1}^n \mathbb{1}_{\{I_t \leq i\}} \mathbb{1}_{\{I_{t-h} \leq i\}}. \tag{A.6}$$

A.5.1 Lemma *Let $X_1, X_2, \dots$ be a stationary ordinal process with corresponding rank counts $I_1, I_2, \dots$, let $h \geq 1$. For $s \leq t$ and $i, j \in \{0, \dots, m-1\}$, it then holds that*

$$\begin{aligned} \text{Cov}[Y_{t;i}, Y_{s;j}] &= f_{ij}^{(t-s)} - f_i f_j, \\ \text{Cov}[Y_{t;m+i}^{(h)}, Y_{s;m+j}^{(h)}] &= P(I_t \leq i, I_{t-h} \leq i, I_s \leq j, I_{s-h} \leq j) - f_{ii}^{(h)} f_{jj}^{(h)}, \\ \text{Cov}[Y_{t;i}, Y_{s;m+j}^{(h)}] &= P(I_t \leq i, I_s \leq j, I_{s-h} \leq j) - f_i f_{jj}^{(h)}, \\ \text{Cov}[Y_{t;m+i}^{(h)}, Y_{s;j}] &= P(I_t \leq i, I_{t-h} \leq i, I_s \leq j) - f_{ii}^{(h)} f_j. \end{aligned}$$

If $X_1, X_2, \dots$ are i. i. d., then these formulae simplify to

$$\begin{aligned}
Cov[Y_{t;i}, Y_{s;j}] &= \begin{cases} f_{\min\{i,j\}} - f_i f_j & \text{if } s = t, \\ 0 & \text{otherwise;} \end{cases} \\
Cov[Y_{t;m+i}^{(h)}, Y_{s;m+j}^{(h)}] &= \begin{cases} f_{\min\{i,j\}}^2 - f_i^2 f_j^2 & \text{if } s = t, \\ f_i f_j f_{\min\{i,j\}} - f_i^2 f_j^2 & \text{if } s = t - h, \\ 0 & \text{otherwise;} \end{cases} \\
Cov[Y_{t;i}, Y_{s;m+j}^{(h)}] &= \begin{cases} f_j f_{\min\{i,j\}} - f_i f_j^2 & \text{if } s = t, \\ 0 & \text{otherwise;} \end{cases} \\
Cov[Y_{t;m+i}^{(h)}, Y_{s;j}] &= \begin{cases} f_i f_{\min\{i,j\}} - f_i^2 f_j & \text{if } s = t, t - h, \\ 0 & \text{otherwise.} \end{cases}
\end{aligned}$$

Proof: We start by computing the cross-covariances between the components of $\mathbf{Y}_t^{(h)}, \mathbf{Y}_s^{(h)}$ with $s \leq t$. For $i, j \in \{0, \dots, m-1\}$, we generally have

$$Cov[Y_{t;i}, Y_{s;j}] = E[\mathbb{1}_{\{I_t \leq i\}} \mathbb{1}_{\{I_s \leq j\}}] - f_i f_j = P(I_t \leq i, I_s \leq j) - f_i f_j.$$

For $t = s$, this simplifies to

$$Cov[Y_{t;i}, Y_{t;j}] = f_{\min\{i,j\}} - f_i f_j.$$

For $t > s$, we generally obtain

$$Cov[Y_{t;i}, Y_{s;j}] = f_{ij}^{(t-s)} - f_i f_j,$$

but in the i. i. d.-case, this covariance just equals 0.

Next, consider the last m components of $\mathbf{Y}_t^{(h)}, \mathbf{Y}_s^{(h)}$ with $s \leq t$. For $i, j \in \{0, \dots, m-1\}$, we have

$$Cov[Y_{t;m+i}^{(h)}, Y_{s;m+j}^{(h)}] = E[\mathbb{1}_{\{I_t \leq i\}} \mathbb{1}_{\{I_{t-h} \leq i\}} \mathbb{1}_{\{I_s \leq j\}} \mathbb{1}_{\{I_{s-h} \leq j\}}] - f_{ii}^{(h)} f_{jj}^{(h)}.$$

In the i. i. d.-case, $f_{ij}^{(h)} = f_i f_j$, and we get

$$\begin{aligned}
Cov[Y_{t;m+i}^{(h)}, Y_{s;m+j}^{(h)}] &= \\
&\begin{cases} E[\mathbb{1}_{\{I_t \leq i\}} \mathbb{1}_{\{I_t \leq j\}}] E[\mathbb{1}_{\{I_{t-h} \leq i\}} \mathbb{1}_{\{I_{s-h} \leq j\}}] - f_i^2 f_j^2 & \text{if } s = t, \\ E[\mathbb{1}_{\{I_t \leq i\}}] E[\mathbb{1}_{\{I_{t-h} \leq i\}} \mathbb{1}_{\{I_{t-h} \leq j\}}] E[\mathbb{1}_{\{I_{t-2h} \leq j\}}] - f_i^2 f_j^2 & \text{if } s = t - h, \\ E[\mathbb{1}_{\{I_t \leq i\}}] E[\mathbb{1}_{\{I_{t-h} \leq i\}}] E[\mathbb{1}_{\{I_s \leq j\}}] E[\mathbb{1}_{\{I_{s-h} \leq j\}}] - f_i^2 f_j^2 & \text{otherwise.} \end{cases}
\end{aligned}$$

So altogether, it follows that

$$\text{Cov}[Y_{t;m+i}^{(h)}, Y_{s;m+j}^{(h)}] = \begin{cases} f_{\min\{i,j\}}^2 - f_i^2 f_j^2 & \text{if } t = s, \\ f_i f_j f_{\min\{i,j\}} - f_i^2 f_j^2 & \text{if } t = s + h, \\ 0 & \text{otherwise.} \end{cases}$$

Finally, again for $s \leq t$ and $i, j \in \{0, \dots, m-1\}$, we consider

$$\begin{aligned} \text{Cov}[Y_{t;i}^{(h)}, Y_{s;m+j}^{(h)}] &= E[\mathbb{1}_{\{I_t \leq i\}} \mathbb{1}_{\{I_s \leq j\}} \mathbb{1}_{\{I_{s-h} \leq j\}}] - f_i f_{jj}^{(h)}, \\ \text{Cov}[Y_{t;m+i}^{(h)}, Y_{s;j}^{(h)}] &= E[\mathbb{1}_{\{I_t \leq i\}} \mathbb{1}_{\{I_{t-h} \leq i\}} \mathbb{1}_{\{I_s \leq j\}}] - f_{ii}^{(h)} f_j. \end{aligned}$$

In the i.i.d.-case, this simplifies to

$$\begin{aligned} \text{Cov}[Y_{t;i}^{(h)}, Y_{s;m+j}^{(h)}] &= \begin{cases} f_j f_{\min\{i,j\}} - f_i f_j^2 & \text{if } s = t, \\ 0 & \text{otherwise;} \end{cases} \\ \text{Cov}[Y_{t;m+i}^{(h)}, Y_{s;j}^{(h)}] &= \begin{cases} f_i f_{\min\{i,j\}} - f_i^2 f_j & \text{if } s = t, t-h, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

So the proof of Lemma A.5.1 is complete. #

If considering marginal properties of the ordinal time series, it suffices to concentrate on the first m components of the vector $\mathbf{Y}_t^{(h)}$ given by (A.5). The cross-covariances corresponding to the univariate empirical CDF are easily computed exactly.

A.5.2 Lemma *Let $X_1, X_2, \dots$ be a stationary ordinal process. For $i, j \in \{0, \dots, m-1\}$, it holds that*

$$\text{Cov}[\hat{f}_i, \hat{f}_j] = \frac{1}{n} (f_{\min\{i,j\}} - f_i f_j) + \frac{1}{n} \sum_{k=1}^{n-1} (1 - \frac{k}{n}) (f_{ij}^{(k)} + f_{ji}^{(k)} - 2 f_i f_j).$$

If $X_1, X_2, \dots$ are i.i.d., then this formula simplifies to

$$\text{Cov}[\hat{f}_i, \hat{f}_j] = \frac{1}{n} (f_{\min\{i,j\}} - f_i f_j).$$

Proof: By (A.6), it follows that

$$\text{Cov}[\hat{f}_i, \hat{f}_j] = \text{Cov}[\hat{f}_i - f_i, \hat{f}_j - f_j] = \frac{1}{n^2} \sum_{t,s=1}^n \text{Cov}[Y_{t;i}^{(h)}, Y_{s;j}^{(h)}].$$

Applying Lemma A.5.1, this leads to

$$\begin{aligned} \text{Cov}[\hat{f}_i, \hat{f}_j] &= \frac{1}{n^2} \sum_{t=1}^n (f_{\min\{i,j\}} - f_i f_j) \\ &\quad + \frac{1}{n^2} \sum_{t>s} (f_{ij}^{(t-s)} - f_i f_j) + \frac{1}{n^2} \sum_{t<s} (f_{ji}^{(s-t)} - f_i f_j). \end{aligned}$$

Using that $\sum_{t=1}^{n-1} \sum_{k=1}^{n-t} = \sum_{k=1}^{n-1} \sum_{t=1}^{n-k}$, this can be rewritten as

$$\begin{aligned} \text{Cov}[\hat{f}_i, \hat{f}_j] &= \frac{1}{n} (f_{\min\{i,j\}} - f_i f_j) + \frac{1}{n^2} \sum_{k=1}^{n-1} (n-k) (f_{ij}^{(k)} - f_i f_j) \\ &\quad + \frac{1}{n^2} \sum_{k=1}^{n-1} (n-k) (f_{ji}^{(k)} - f_i f_j). \end{aligned}$$

The proof is completed by considering that the last two summands become 0 in the i.i.d.-case. #

For deriving the asymptotic distribution of the considered ordinal measures, we make use of an appropriate central limit theorem (CLT) and a subsequent application of the Delta method. A CLT is applicable to $(\mathbf{Y}_t^{(h)})_{\mathbb{Z}}$ according to (A.5) if the ordinal process satisfies a sufficient mixing condition (Bradley, 2005). A rather weak mixing condition, but which is often satisfied in practice, is as follows.

A.5.3 Definition Let $(X_t)_{\mathbb{Z}}$ be a stationary process on the probability space $(\Omega, \mathfrak{A}, P)$. It is said to be α -mixing if there exists a non-negative sequence $(\alpha_n)_{\mathbb{N}}$ of weights with $\alpha_n \rightarrow 0$ for $n \rightarrow \infty$ such that for each $t \in \mathbb{Z}$, $k \in \mathbb{N}$ and all events $E_1 \in \mathfrak{A}(X_t, X_{t-1}, \dots)$, $E_2 \in \mathfrak{A}(X_{t+k}, X_{t+k+1}, \dots)$, the following inequality holds:

$$|P(E_1 \cap E_2) - P(E_1)P(E_2)| \leq \alpha_k.$$

If the ordinal process is α -mixing with exponentially decreasing weights (i.e., the α_k in Definition A.5.3 are of the form $c\rho^k$ with $\rho \in [0; 1)$), then the CLT in Ibragimov (1962) can be applied to $(\mathbf{Y}_t^{(h)})_{\mathbb{Z}}$, and it follows that

$$\begin{aligned} \frac{1}{\sqrt{n}} \sum_{t=1}^n \mathbf{Y}_t^{(h)} &\xrightarrow{d} N(\mathbf{0}, \Sigma^{(h)}) \quad \text{with } \Sigma^{(h)} = (\sigma_{i,j}^{(h)})_{i,j=0,\dots,2m-1}, \quad \text{where} \\ \sigma_{i,j}^{(h)} &= E[Y_{0,i}^{(h)} Y_{0,j}^{(h)}] + \sum_{t=1}^{\infty} \left(E[Y_{0,i}^{(h)} Y_{t,j}^{(h)}] + E[Y_{t,i}^{(h)} Y_{0,j}^{(h)}] \right). \end{aligned} \tag{A.7}$$

The required expressions for $E[Y_{t,i}^{(h)} Y_{s,j}^{(h)}] = \text{Cov}[Y_{t,i}^{(h)}, Y_{s,j}^{(h)}]$ in (A.7) are provided by Lemma A.5.1. In particular, the covariances $\sigma_{i,j} = \sigma_{i,j}^{(h)}$ for $i, j \in \{0, \dots, m-1\}$ are approximated as

$$\sigma_{i,j} \stackrel{a}{=} f_{\min\{i,j\}} - f_i f_j + \sum_{k=1}^{\infty} (f_{ij}^{(k)} + f_{ji}^{(k)} - 2 f_i f_j), \tag{A.8}$$

whereas the exact expressions are provided by Lemma A.5.2. In the i. i. d.-case, the required mixing condition is obviously satisfied. For the covariances in (A.7), it then follows for $i, j \in \{0, \dots, m-1\}$ that

$$\begin{aligned}\sigma_{i,j} &= f_{\min\{i,j\}} - f_i f_j, \\ \sigma_{i,m+j}^{(h)} &= 2 f_j (f_{\min\{i,j\}} - f_i f_j), \\ \sigma_{m+i,m+j}^{(h)} &= f_{\min\{i,j\}}^2 - f_i^2 f_j^2 + 2 f_i f_j (f_{\min\{i,j\}} - f_i f_j) \\ &= (f_{\min\{i,j\}} + 3 f_i f_j) (f_{\min\{i,j\}} - f_i f_j).\end{aligned}\tag{A.9}$$

A.6 Derivations of Sections 4 and 7.1

Estimation of $E[I] = m - \sum_{i=0}^{m-1} f_i$ is relevant for quantifying the skewness of the ordinal marginal distribution, see (13). As the estimator, we consider the corresponding sample mean $\bar{I} = m - \sum_{i=0}^{m-1} \hat{f}_i$. The variance of this unbiased and asymptotically normally distributed estimator is obtained in two ways, which is exemplified for the i. i. d.-case: On the one hand, it is well known that

$$V[\bar{I}] \stackrel{\text{i.i.d.}}{=} \frac{1}{n} V[I] = \frac{1}{n} \sum_{i=0}^{m-1} f_i (1 - \sum_{j=0}^{m-1} f_j) + \frac{2}{n} \sum_{i=1}^{m-1} \sum_{j=0}^{i-1} f_j,$$

where the last equality follows from (A.3). On the other hand, using Lemma A.5.2, we also compute

$$V[\bar{I}] = V[\sum_{i=0}^{m-1} \hat{f}_i] = \sum_{i,j=0}^{m-1} \text{Cov}[\hat{f}_i, \hat{f}_j] \stackrel{\text{i.i.d.}}{=} \frac{1}{n} \sum_{i,j=0}^{m-1} (f_{\min\{i,j\}} - f_i f_j).$$

Note that these two derivations are easily modified to the non-i. i. d.-case by considering either the rank counts' ACF $\rho_I(k)$, i. e.,

$$V[\bar{I}] = \frac{1}{n} V[I] (1 + 2 \sum_{k=1}^{n-1} (1 - \frac{k}{n}) \rho_I(k)),$$

or the general expressions for $\text{Cov}[\hat{f}_i, \hat{f}_j]$ in Lemma A.5.2.

As a consequence, the sample version (15) of the skewness measure (13) is unbiased with i. i. d.-variance

$$V[\widehat{\text{skew}}_{d_o,1}] = 4 V[\bar{I}] = \frac{4}{n} \sum_{i,j=0}^{m-1} (f_{\min\{i,j\}} - f_i f_j),$$

which completes the proof of part (iii) in Theorems 4.1 and 7.1.1.

Asymptotic distributions for the sample dispersion measure $\widehat{\text{disp}}_d$ and asymmetry measure $\widehat{\text{asym}}_d$ could be derived by using the quadratic form representations of these measures in (14) and (15). The asymptotic normality of $\sqrt{n}\hat{\mathbf{p}}$ together with Theorem 3.1 in Tan (1977) implies an asymptotic quadratic form distribution for these measures. In general, however, the non-centrality terms of this distribution will not vanish such that the computation of the CDF or quantiles is demanding (Duchesne & de Micheaux, 2010). Therefore, it appears to be more convenient for practice to use again an asymptotic normal distribution to approximate the true distributions of $\widehat{\text{disp}}_d$ and $\widehat{\text{asym}}_d$, respectively.

The sample version of the dispersion measure $\text{disp}_{d_{o,1}}$, see (7), is given by $\widehat{\text{disp}}_{d_{o,1}} = 2 \sum_{i=0}^{m-1} \hat{f}_i(1 - \hat{f}_i)$. It is asymptotically normally distributed as implied by (A.7) and the Delta method. Since

$$E[\hat{f}_i(1 - \hat{f}_i)] = E[\hat{f}_i] - E[\hat{f}_i^2] = f_i(1 - f_i) - V[\hat{f}_i],$$

it follows that $\widehat{\text{disp}}_{d_{o,1}}$ always has a negative bias. The exact bias can be computed by using Lemma A.5.2:

$$E[\hat{f}_i(1 - \hat{f}_i)] = (1 - \frac{1}{n}) f_i(1 - f_i) - \frac{2}{n} \sum_{k=1}^{n-1} (1 - \frac{k}{n}) (f_{ii}^{(k)} - f_i^2).$$

In the i. i. d.-case, it follows that $E[\widehat{\text{disp}}_{d_{o,1}}] = (1 - \frac{1}{n}) \text{disp}_{d_{o,1}}$, also see Kiesl (2003), which is in agreement with the corrective factor $\frac{n}{n-1}$ in (14). For the asymptotic variance, we use the Delta method together with the asymptotic result (A.7). Define the function $g(\mathbf{x}) = 2 \sum_{i=0}^{m-1} x_i(1 - x_i)$, having the partial derivatives $\frac{\partial}{\partial x_i} g(\mathbf{x}) = 2(1 - 2x_i)$. If the ordinal distribution is not the extreme two-point distribution, then $\mathbf{D} = \text{grad } g(f_0, \dots, f_{m-1})$ is not identical to $(0, \dots, 0)$, i. e., the linear approximation $\text{disp}_{d_{o,1}} + \mathbf{D}(\hat{\mathbf{f}} - \mathbf{f})$ of $\widehat{\text{disp}}_{d_{o,1}}$ does not reduce to the constant value $\text{disp}_{d_{o,1}}$. We obtain that

$$V[\widehat{\text{disp}}_{d_{o,1}}] \stackrel{\text{a}}{=} \frac{1}{n} \mathbf{D} \boldsymbol{\Sigma} \mathbf{D}^\top = \frac{4}{n} \sum_{i,j=0}^{m-1} (1 - 2f_i)(1 - 2f_j) \sigma_{i,j},$$

where $\boldsymbol{\Sigma} = (\sigma_{i,j})_{i,j=0,\dots,m-1}$ according to (A.8). In the i. i. d.-case, (A.9) implies that

$$\begin{aligned} V[\widehat{\text{disp}}_{d_{o,1}}] &\stackrel{\text{a}}{=} \frac{4}{n} \sum_{i,j=0}^{m-1} (1 - 2f_i)(1 - 2f_j) (f_{\min\{i,j\}} - f_i f_j) \\ &= \frac{4}{n} \sum_{i=0}^{m-1} (1 - 2f_i)^2 f_i(1 - f_i) + \frac{8}{n} \sum_{i < j} (1 - 2f_i)(1 - 2f_j) f_i(1 - f_j), \end{aligned}$$

where the last expression was given in Kiesl (2003). Altogether, the proof of part (i) in Theorems 4.1 and 7.1.1 is complete.

In the case of an extreme two-point distribution, see Remarks 4.2 and 7.1.2, the linear term in the Taylor approximation vanishes, and $\widehat{\text{IOV}} = 1$. In this case, however, we necessarily have $\hat{f}_0 = \dots = \hat{f}_{m-1}$ such that $\widehat{\text{IOV}} = 4\hat{f}_0(1 - \hat{f}_0) = 1 - 4(\hat{f}_0 - \frac{1}{2})^2$. From Lemma A.5.2 and (A.7), we know that $\sqrt{n}(\hat{f}_0 - \frac{1}{2})$ is asymptotically distributed like $N(0, \sigma_{0,0})$ such that $n(\widehat{\text{IOV}} - 1)$ asymptotically follows $-4\sigma_{0,0}Z^2$ with Z being a $N(0, 1)$ -variate. So asymptotic mean and variance immediately follow from $E[Z^2] = 1$ and $V[Z^2] = 2$, and quantiles of $\widehat{\text{IOV}}$ are approximated by using those of a χ_1^2 -distribution.

Finally, we turn to the sample asymmetry measure $\widehat{\text{asym}}_{d_{o,1}} = \sum_{j=0}^{m-1} (1 - \hat{f}_j - \hat{f}_{m-j-1})^2$, see (11). In analogy to our computations for $\widehat{\text{IOV}}$, it is easily seen that $\widehat{\text{asym}}_{d_{o,1}}$ is positively biased, since

$$\begin{aligned} E[(1 - \hat{f}_j - \hat{f}_{m-j-1})^2] &= E[1 - \hat{f}_j - \hat{f}_{m-j-1}]^2 + V[1 - \hat{f}_j - \hat{f}_{m-j-1}] \\ &= (1 - f_j - f_{m-j-1})^2 + V[\hat{f}_j] + V[\hat{f}_{m-j-1}] + 2\text{Cov}[\hat{f}_j, \hat{f}_{m-j-1}], \end{aligned}$$

where the latter terms can be computed using Lemma A.5.2. In the i.i.d.-case, we end up with

$$E[\widehat{\text{asym}}_{d_{o,1}}] = \text{asym}_{d_{o,1}} + \frac{2}{n} \sum_{j=0}^{m-1} \left(f_j(1 - f_j) + f_{\min\{j, m-j-1\}} - f_j f_{m-j-1} \right).$$

For the asymptotic variance, we use the Delta method together with the asymptotic result (A.7). Define the function $g(\mathbf{x}) = \sum_{k=0}^{m-1} (1 - x_k - x_{m-k-1})^2$, having the partial derivatives $\frac{\partial}{\partial x_k} g(\mathbf{x}) = -4(1 - x_k - x_{m-k-1})$ (this expression also holds if $k = \frac{m-1}{2}$). If the ordinal distribution is not symmetric, then $\mathbf{D} = \text{grad } g(f_0, \dots, f_{m-1})$ is not identical to $(0, \dots, 0)$, i.e., the linear approximation $\widehat{\text{asym}}_{d_{o,1}} + \mathbf{D}(\mathbf{f} - \mathbf{f})$ of $\widehat{\text{asym}}_{d_{o,1}}$ does not reduce to the constant value $\text{asym}_{d_{o,1}}$. We obtain that

$$V[\widehat{\text{asym}}_{d_{o,1}}] \stackrel{\text{a}}{=} \frac{1}{n} \mathbf{D} \mathbf{\Sigma} \mathbf{D}^\top = \frac{16}{n} \sum_{i,j=0}^{m-1} (1 - f_i - f_{m-i-1})(1 - f_j - f_{m-j-1}) \sigma_{i,j},$$

where $\mathbf{\Sigma} = (\sigma_{i,j})_{i,j=0,\dots,m-1}$ according to (A.8). In the i.i.d.-case, (A.9) implies that

$$V[\widehat{\text{asym}}_{d_{o,1}}] \stackrel{\text{a}}{=} \frac{16}{n} \sum_{i,j=0}^{m-1} (1 - f_i - f_{m-i-1})(1 - f_j - f_{m-j-1}) (f_{\min\{i,j\}} - f_i f_j),$$

which completes the proof of part (ii) in Theorems 4.1 and 7.1.1.

In the case of a symmetric distribution, see Remarks 4.2 and 7.1.2, the linear term in the Taylor approximation vanishes (and $\text{asym}_{d_{o,1}} = 0$) such that a quadratic approximation of

$\widehat{\text{asym}}_{d_{o,1}}$ is required, i. e., $0 + \frac{1}{2} (\hat{\mathbf{f}} - \mathbf{f})^\top \mathbf{H} (\hat{\mathbf{f}} - \mathbf{f})$, where $\mathbf{H}$ denotes the Hessian of g , evaluated in $(f_0, \dots, f_{m-1})$. It is easily seen that the Hessian of g just equals $4(\mathbf{I} + \mathbf{J})$, where $\mathbf{I}$ ($\mathbf{J}$) denotes the m -dimensional identity matrix (exchange matrix). So Theorem 3.1 in Tan (1977), together with the asymptotic normality of $\sqrt{n} \hat{\mathbf{f}}$, implies the following approximate quadratic form distribution for $\widehat{\text{asym}}_{d_{o,1}}$ in the symmetry case: Let $\lambda_1, \dots, \lambda_u$ denote the non-zero eigenvalues of $\Sigma(\mathbf{I} + \mathbf{J})$, where Σ is given by (A.8). Then the distribution of $n \widehat{\text{asym}}_{d_{o,1}}$ is approximated by the one of the quadratic form $2 \sum_{i=1}^u \lambda_i Z_i^2$ with $Z_1, \dots, Z_u$ being i. i. d. $N(0, 1)$ -variates. Asymptotic mean and variance follow by considering that each Z_i^2 has mean 1 and variance 2.

A.7 Derivations of Section 7.2

Applying equation (19), also see Supplement A.4, we immediately obtain for $(I_t)_{\mathbb{Z}}$ being stationary that

$$E[|I_t - I_{t-h}|] = 2 \sum_{k=0}^{m-1} \left(P(I_t \leq k) - P(I_t \leq k, I_{t-h} \leq k) \right) = 2 \sum_{k=0}^{m-1} (f_k - f_{kk}^{(h)}).$$

Note that with the additional assumption of pairwise independence, we would obtain

$$E[|I_t - I_{t-h}|] = 2 \sum_{k=0}^{m-1} f_k(1 - f_k) = \frac{m}{2} - \frac{1}{2} \sum_{k=0}^{m-1} (2f_k - 1)^2,$$

also see (7). Plugging-in into (21), we end up with

$$\kappa_{d_{o,1}}(h) = \frac{\sum_{k=0}^{m-1} (f_{kk}^{(h)} - f_k^2)}{\sum_{k=0}^{m-1} f_k(1 - f_k)},$$

so the proof of equation (23) is complete.

Next, we consider the sample version $\hat{\kappa}_{d_{o,1}}(h)$ of $\kappa_{d_{o,1}}(h)$. Its distribution is approximated by the asymptotic distribution of

$$g\left(\frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t^{(h)}\right), \quad \text{where } g(\mathbf{x}) = \frac{\sum_{i=0}^{m-1} (x_{m+i} - x_i^2)}{\sum_{i=0}^{m-1} x_i(1 - x_i)} = \frac{\sum_{i=0}^{m-1} (x_{m+i} - x_i)}{\sum_{i=0}^{m-1} x_i(1 - x_i)} + 1;$$

see (A.4) for the definition of $\mathbf{Z}_t^{(h)}$. The partial derivatives of g are computed as

$$\frac{\partial}{\partial x_{m+j}} g(\mathbf{x}) = \frac{1}{\sum_{i=0}^{m-1} x_i(1 - x_i)}, \quad \frac{\partial}{\partial x_j} g(\mathbf{x}) = \frac{-2x_j - (1 - 2x_j)g(\mathbf{x})}{\sum_{i=0}^{m-1} x_i(1 - x_i)}. \quad (\text{A.10})$$

Thus, the gradient of g , evaluated in the $f_i, f_{ii}^{(h)}$, becomes

$$\mathbf{D} = \frac{1}{\sum_{i=0}^{m-1} f_i(1-f_i)} \left(\dots, -2f_j - (1-2f_j) \kappa_{d_{o,1}}(h), \dots, 1, \dots, 1 \right).$$

So applying the Delta method to the asymptotic result (A.7), the asymptotic variance of $\sqrt{n} g\left(\frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t^{(h)}\right)$ becomes

$$\begin{aligned} \mathbf{D} \Sigma^{(h)} \mathbf{D}^\top &= \frac{1}{\left(\sum_{i=0}^{m-1} f_i(1-f_i)\right)^2} \sum_{k,l=0}^{m-1} \left(-2(2f_k + (1-2f_k) \kappa_{d_{o,1}}(h)) \sigma_{k,m+l}^{(h)} \right. \\ &\quad \left. + \sigma_{m+k,m+l}^{(h)} + (2f_k + (1-2f_k) \kappa_{d_{o,1}}(h)) (2f_l + (1-2f_l) \kappa_{d_{o,1}}(h)) \sigma_{k,l}^{(h)} \right). \end{aligned}$$

In the i. i. d.-case, this expression further simplifies by using (A.9) and by considering that $\kappa_{d_{o,1}}(h) = 0$:

$$\begin{aligned} \mathbf{D} \Sigma^{(h)} \mathbf{D}^\top &= \frac{1}{\left(\sum_{i=0}^{m-1} f_i(1-f_i)\right)^2} \sum_{k,l=0}^{m-1} \left(4f_k f_l \sigma_{k,l}^{(h)} - 4f_k \sigma_{k,m+l}^{(h)} + \sigma_{m+k,m+l}^{(h)} \right) \\ &= \frac{1}{\left(\sum_{i=0}^{m-1} f_i(1-f_i)\right)^2} \sum_{k,l=0}^{m-1} (f_{\min\{k,l\}} - f_k f_l) \left(4f_k f_l - 8f_k f_l + (f_{\min\{k,l\}} + 3f_k f_l) \right) \\ &= \frac{1}{\left(\sum_{i=0}^{m-1} f_i(1-f_i)\right)^2} \sum_{k,l=0}^{m-1} (f_{\min\{k,l\}} - f_k f_l)^2. \end{aligned}$$

To derive an approximate bias correction, we consider a second-order Taylor approximation of $g(\mathbf{x})$. Using (A.10), we compute the following second-order partial derivatives for $j, k = 0, \dots, m-1$:

$$\begin{aligned} \frac{\partial^2}{\partial x_{m+k} \partial x_{m+j}} g(\mathbf{x}) &= 0, \quad \frac{\partial^2}{\partial x_k \partial x_{m+j}} g(\mathbf{x}) = \frac{-(1-2x_k)}{\left(\sum_{i=0}^{m-1} x_i(1-x_i)\right)^2}, \\ \frac{\partial^2}{\partial x_k \partial x_j} g(\mathbf{x}) &= 2 \frac{x_j + x_k - 4x_j x_k + (1-2x_j)(1-2x_k) g(\mathbf{x})}{\left(\sum_{i=0}^{m-1} x_i(1-x_i)\right)^2} - \frac{\delta_{k,j} 2(1-g(\mathbf{x}))}{\sum_{i=0}^{m-1} x_i(1-x_i)}. \end{aligned} \tag{A.11}$$

Here, the last derivative follows from

$$\begin{aligned}
\frac{\partial^2}{\partial x_k \partial x_j} g(\mathbf{x}) &\stackrel{(A.10)}{=} \frac{\partial}{\partial x_k} \frac{-2x_j - (1-2x_j)g(\mathbf{x})}{\sum_{i=0}^{m-1} x_i(1-x_i)} \\
&= \frac{(2x_j + (1-2x_j)g(\mathbf{x}))(1-2x_k)}{(\sum_{i=0}^{m-1} x_i(1-x_i))^2} - \frac{\frac{\partial}{\partial x_k} (2x_j + (1-2x_j)g(\mathbf{x}))}{\sum_{i=0}^{m-1} x_i(1-x_i)} \\
&= \frac{(2x_j + (1-2x_j)g(\mathbf{x}))(1-2x_k)}{(\sum_{i=0}^{m-1} x_i(1-x_i))^2} - \frac{\delta_{k,j}(2-2g(\mathbf{x})) + (1-2x_j)\frac{\partial}{\partial x_k} g(\mathbf{x})}{\sum_{i=0}^{m-1} x_i(1-x_i)} \\
&\stackrel{(A.10)}{=} \frac{(2x_j + (1-2x_j)g(\mathbf{x}))(1-2x_k)}{(\sum_{i=0}^{m-1} x_i(1-x_i))^2} - \frac{\delta_{k,j}2(1-g(\mathbf{x}))}{\sum_{i=0}^{m-1} x_i(1-x_i)} \\
&\quad + \frac{(1-2x_j)(2x_k + (1-2x_k)g(\mathbf{x}))}{(\sum_{i=0}^{m-1} x_i(1-x_i))^2} \\
&= 2 \frac{x_j + x_k - 4x_j x_k + (1-2x_j)(1-2x_k)g(\mathbf{x})}{(\sum_{i=0}^{m-1} x_i(1-x_i))^2} - \frac{\delta_{k,j}2(1-g(\mathbf{x}))}{\sum_{i=0}^{m-1} x_i(1-x_i)}.
\end{aligned}$$

Using (A.11), we compute the matrix $\mathbf{H}$, which is defined to be the Hessian of g evaluated in the $f_i, f_{ii}^{(h)}$. In the i.i.d.-case, where $\kappa_{d_{o,1}}(h) = 0$, this matrix has the entries

$$\begin{aligned}
h_{m+k,m+j} &= 0, \quad h_{k,m+j} = \frac{-(1-2f_k)}{(\sum_{i=0}^{m-1} f_i(1-f_i))^2}, \\
h_{k,j} &= 2 \frac{f_j + f_k - 4f_j f_k}{(\sum_{i=0}^{m-1} f_i(1-f_i))^2} - \frac{\delta_{k,j}2}{\sum_{i=0}^{m-1} f_i(1-f_i)}.
\end{aligned}$$

Then a bias correction for $n \hat{\kappa}_{d_{o,1}}(h)$ is computed as

$$\begin{aligned}
&\frac{1}{2} \sum_{k=0}^{2m-1} h_{k,k} \sigma_{k,k}^{(h)} + \sum_{k=0}^{2m-2} \sum_{j=k+1}^{2m-1} h_{k,j} \sigma_{k,j}^{(h)} \\
&= \frac{1}{2} \sum_{k=0}^{m-1} h_{k,k} \sigma_{k,k}^{(h)} + \sum_{k=0}^{m-2} \sum_{j=k+1}^{m-1} h_{k,j} \sigma_{k,j}^{(h)} + \sum_{k,j=0}^{m-1} h_{k,m+j} \sigma_{k,m+j}^{(h)} \\
&= \frac{1}{2} \sum_{k,j=0}^{m-1} h_{k,j} \sigma_{k,j}^{(h)} + \sum_{k,j=0}^{m-1} h_{k,m+j} \sigma_{k,m+j}^{(h)} \\
&= - \frac{\sum_{k=0}^{m-1} \sigma_{k,k}^{(h)}}{\sum_{i=0}^{m-1} f_i(1-f_i)} + \frac{\sum_{k,j=0}^{m-1} (f_j + f_k - 4f_j f_k) \sigma_{k,j}^{(h)}}{(\sum_{i=0}^{m-1} f_i(1-f_i))^2} - \frac{\sum_{k,j=0}^{m-1} (1-2f_k) \sigma_{k,m+j}^{(h)}}{(\sum_{i=0}^{m-1} f_i(1-f_i))^2} \\
&\stackrel{(A.9)}{=} -1 + \frac{\sum_{k,j=0}^{m-1} (f_j + f_k - 4f_j f_k)(f_{\min\{k,j\}} - f_k f_j)}{(\sum_{i=0}^{m-1} f_i(1-f_i))^2} - 2 \frac{\sum_{k,j=0}^{m-1} f_j(1-2f_k)(f_{\min\{k,j\}} - f_k f_j)}{(\sum_{i=0}^{m-1} f_i(1-f_i))^2},
\end{aligned}$$

which just reduces to -1 . So the approximate bias of $\hat{\kappa}_{d_{o,1}}(h)$ equals $-1/n$ in the i.i.d.-case.

B Numerical Illustrations

Figure 1 compares the (normalized) dispersion measures (6), (7) and (8) for $m = 4$ (a common choice for Likert scales), where the rank count follows either the binomial distribution $\text{Bin}(m, \pi)$ or the zero-inflated binomial distribution $\text{ZIB}(m, \pi, \omega)$. In part (a) referring to the binomial situation, all dispersion measures become maximal for $\pi = 0.5$, but their absolute amounts differ considerably, since $\text{Bin}(4, 0.5)$ is more close to a uniform distribution than to an extreme two-point distribution. The ZIB distributions considered in part (b) are parametrized such that their mean is always the same, namely $E[I] = 0.5m = 2$, but the variance increases with increasing ω , where $\omega = 0$ corresponds to the binomial variance $0.25m = 1$. This extra-binomial variation is caused by zero inflation: the zero probability $\omega + (1 - \omega)0.5^m$ increases with increasing ω , whereas the probability mass associated with the distribution $\text{Bin}(4, \frac{0.5}{1-\omega})$ moves towards larger values. For $\omega \rightarrow 0.5$, the $\text{ZIB}(4, \frac{0.5}{1-\omega}, \omega)$ distribution converges to the extreme two-point distribution such that $\text{disp}_{d_{o,1}}, \text{disp}_{d_{o,2}} \rightarrow 1$. The measure disp_{d_H} , in contrast, becomes maximal for $\omega \approx 0.198$ where $\text{ZIB}(4, \frac{0.5}{1-\omega}, \omega)$ is relatively close to the uniform distribution.

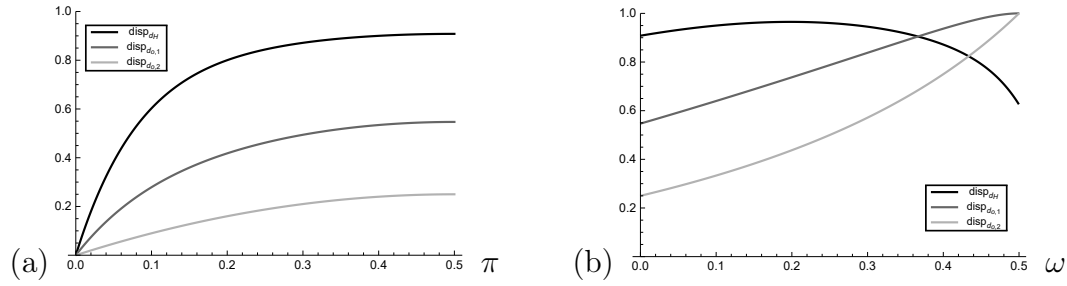


Figure 1: Normalized version of dispersion measures disp_{d_H} , $\text{disp}_{d_{o,1}}$, $\text{disp}_{d_{o,2}}$ for rank count I distributed as (a) $\text{Bin}(4, \pi)$, (b) $\text{ZIB}(4, \frac{0.5}{1-\omega}, \omega)$.

Figure 2 compares the (normalized) asymmetry measures asym_{d_H} , $\text{asym}_{d_{o,1}}$ and skewness measures skew_{d_H} , $\text{skew}_{d_{o,1}}$ for $m = 4$. The rank count's distribution $\text{Bin}(4, \pi)$ is symmetric for $\pi = 0.5$ and asymmetric otherwise, as recognized by all measures in part (a). Both skewness measures also recognize the positive (negative) skewness for $\pi < 0.5$ ($\pi > 0.5$). The plot in (b) starts with the symmetric $\text{Bin}(4, 0.5)$ distribution for $\omega \rightarrow 0$, and it ends with the symmetric extreme two-point distribution for $\omega \rightarrow 1$. For $\omega \in (0; 1)$, in contrast, the distribution $\text{ZIB}(4, \frac{0.5}{1-\omega}, \omega)$ is truly asymmetric. The measure $\text{skew}_{d_{o,1}}$ from (13), however, does not recognize this asymmetry since we constantly have $E[I] = 2 = \frac{m}{2}$. The remaining measures, in contrast, lead to varying asymmetry and skewness values.

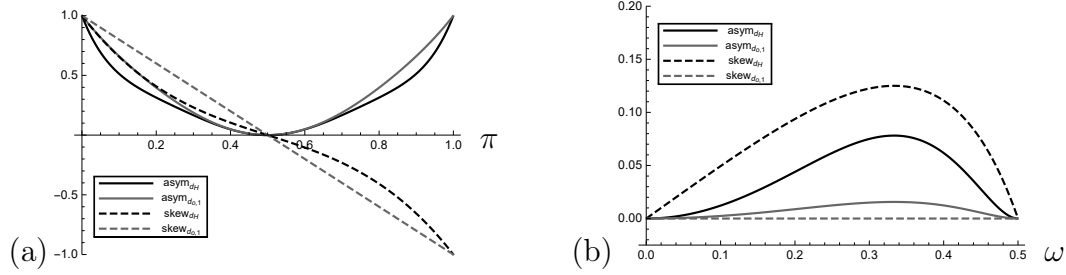


Figure 2: Normalized version of asymmetry measures asym_{d_H} , $\text{asym}_{d_{o,1}}$ and skewness measures skew_{d_H} , $\text{skew}_{d_{o,1}}$ for rank count I distributed as (a) $\text{Bin}(4, \pi)$, (b) $\text{ZIB}(4, \frac{0.5}{1-\omega}, \omega)$.

C Simulations

C.1 Dispersion and Symmetry for i.i.d. Ordinal Samples

The asymptotic distributions for the sample measures relying on $d_{o,1}$ derived in Section 4 can be used in practice as approximations to their true distributions. The resulting quality of approximation was investigated with simulations (always 10 000 replications per scenario). For a given scenario, the simulated statistics' standard deviation was compared with the corresponding asymptotic values according to Section 4 (the mean formulae given there are exact anyway). Furthermore, the asymptotic normal distributions were also used to construct a simple plug-in confidence interval (level 95 %), i. e., the asymptotic formulae were used together with the respective sample CDF $\hat{f}$. Below, the resulting coverage rates are reported. To keep the article at an acceptable length, only a brief summary of these simulations is presented here (with a few illustrative results), but the full simulation results are available from the author upon request.

Table C.1: Standard deviation (s.d.), asymptotic (-a) vs. simulated (-s), as well as simulated coverage rate for normalized sample measures $\widehat{\text{disp}}_{d_{o,1}}$, $\widehat{\text{asym}}_{d_{o,1}}$ and $\widehat{\text{skew}}_{d_{o,1}}$; i. i. d. rank counts.

n	Bin(4, 0.3) $\frac{2}{m} \text{disp}_{d_{o,1}} = 0.494$			Bin(4, 0.3) $\frac{1}{m} \text{asym}_{d_{o,1}} = 0.188$			Bin(4, 0.3) $\frac{1}{m} \text{skew}_{d_{o,1}} = 0.400$			ZIB(4, $\frac{0.50}{1-0.45}, 0.45$) $\frac{2}{m} \text{disp}_{d_{o,1}} = 0.979$			Bin(4, 0.45) $\frac{1}{m} \text{asym}_{d_{o,1}} = 0.012$		
	s.d.-a	s.d.-s	cov.	s.d.-a	s.d.-s	cov.	s.d.-a	s.d.-s	cov.	s.d.-a	s.d.-s	cov.	s.d.-a	s.d.-s	cov.
50	0.051	0.051	0.930	0.059	0.059	0.922	0.065	0.065	0.938	0.013	0.028	0.901	0.018	0.020	0.797
100	0.036	0.036	0.942	0.042	0.042	0.936	0.046	0.046	0.941	0.009	0.015	0.927	0.013	0.013	0.840
250	0.023	0.022	0.951	0.026	0.026	0.946	0.029	0.029	0.947	0.006	0.008	0.946	0.008	0.008	0.897
500	0.016	0.016	0.947	0.019	0.019	0.951	0.020	0.020	0.953	0.004	0.005	0.951	0.006	0.006	0.916
1000	0.011	0.011	0.949	0.013	0.013	0.951	0.014	0.014	0.951	0.003	0.003	0.954	0.004	0.004	0.937

In most cases, the asymptotic normal approximations according to Theorem 4.1 perform rather well. This is illustrated by the first three blocks in Table C.1, where we see a nearly perfect agreement between asymptotic and simulated standard deviation for all sample sizes n . Also the coverage rates are quite close to the intended confidence level of 95 %. The performance only becomes worse if the considered scenario is close to the excluded scenarios (extreme two-point distribution for $\widehat{\text{disp}}_{d_{o,1}}$, and symmetric distribution for $\widehat{\text{asym}}_{d_{o,1}}$), see the last two blocks, then rather large sample sizes are required.

The asymptotics for these special cases are provided by Remark 4.2, and simulations again confirm a very good finite-sample performance. For illustration, Tables C.2 and C.3

Table C.2: Simulated rejection rates for symmetry tests (level 5 %) relying on either $\widehat{\text{asym}}_{d_{o,1}}$ or $\widehat{\text{skew}}_{d_{o,1}}$; i. i. d. Bin(4, π) rank counts.

n	$\widehat{\text{asym}}_{d_{o,1}}; \pi =$						$\widehat{\text{skew}}_{d_{o,1}}; \pi =$					
	0.50	0.48	0.46	0.44	0.42	0.40	0.50	0.48	0.46	0.44	0.42	0.40
50	0.049	0.083	0.205	0.368	0.581	0.786	0.046	0.084	0.206	0.384	0.602	0.802
100	0.051	0.120	0.340	0.647	0.875	0.978	0.052	0.127	0.354	0.667	0.890	0.982
250	0.050	0.235	0.697	0.961	0.999	1.000	0.049	0.245	0.713	0.966	0.999	1.000
500	0.050	0.407	0.942	1.000	1.000	1.000	0.048	0.424	0.951	1.000	1.000	1.000
1000	0.050	0.704	0.999	1.000	1.000	1.000	0.047	0.718	0.999	1.000	1.000	1.000

Table C.3: Simulated rejection rates for symmetry tests (level 5 %) relying on either $\widehat{\text{asym}}_{d_{o,1}}$ or $\widehat{\text{skew}}_{d_{o,1}}$; i. i. d. ZIB(4, $\frac{0.5}{1-\omega}, \omega$) rank counts.

n	$\widehat{\text{asym}}_{d_{o,1}}; \omega =$					$\widehat{\text{skew}}_{d_{o,1}}; \omega =$				
	0.00	0.05	0.10	0.15	0.20	0.00	0.05	0.10	0.15	0.20
50	0.050	0.051	0.066	0.082	0.105	0.048	0.045	0.048	0.049	0.050
100	0.051	0.056	0.090	0.143	0.222	0.052	0.048	0.050	0.048	0.051
250	0.050	0.076	0.188	0.469	0.782	0.053	0.048	0.048	0.053	0.049
500	0.049	0.116	0.466	0.923	0.998	0.046	0.051	0.046	0.049	0.052
1000	0.055	0.219	0.918	1.000	1.000	0.053	0.048	0.052	0.050	0.048

present rejection rates if $\widehat{\text{asym}}_{d_{o,1}}$ is used to test the null hypothesis of symmetry (for comparison, also the corresponding results according to the $\widehat{\text{skew}}_{d_{o,1}}$ -test are shown). The tests are again implemented by plugging-in an estimate of the CDF into the respective asymptotic distribution under the null. However, since the ordinal distribution is symmetric under the null, it appears to be plausible to plug-in the symmetrized estimate obtained by computing $\hat{p}_{j;\text{sym}} = (\hat{p}_j + \hat{p}_{m-j})/2$. The resulting size and power values are shown in Tables C.2 and C.3; if using the ordinary sample CDF for plug-in instead, then the power values slightly increase at the cost of increased sizes. The sizes shown in Tables C.2 and C.3 are close to the intended 5 %-level. In the binomial situation of Table C.2, where asymmetry goes along with a change in the rank counts' mean, the $\widehat{\text{skew}}_{d_{o,1}}$ -test has a superior power, especially for small n . But in the ZIB-situation of Table C.3, also remember the discussion of Figure 2, the $\widehat{\text{skew}}_{d_{o,1}}$ -test does not show any power whereas the $\widehat{\text{asym}}_{d_{o,1}}$ -test is still sensitive to this kind of asymmetry.

C.2 Dispersion and Symmetry for Ordinal Time Series

The finite-sample performance of the asymptotics provided by Theorem 7.1.1 is investigated in Table C.4, which extends Table C.1 on the i.i.d.-case ($\rho = 0$). The rank counts are now generated by a so-called binomial autoregressive process of order 1 (Bin-AR(1)) with marginal distribution $\text{Bin}(4, 0.3)$ and autocorrelation parameter $\rho = \rho_I(1)$; see, for example, Weiß (2018) for details on this model. From Table C.4, we can see that the approximation quality suffers from increasing ρ such that larger sample sizes are required if dealing with time series data.

Table C.4: Standard deviation (s.d.), asymptotic (-a) vs. simulated (-s), as well as simulated coverage rate for normalized sample measures $\widehat{\text{disp}}_{d_{o,1}}$, $\widehat{\text{asym}}_{d_{o,1}}$ and $\widehat{\text{skew}}_{d_{o,1}}$; $\text{Bin}(4, 0.3)$ -AR(1) rank counts with autocorrelation parameter ρ .

ρ	n	$\frac{2}{m} \text{disp}_{d_{o,1}} = 0.494$			$\frac{1}{m} \text{asym}_{d_{o,1}} = 0.188$			$\frac{1}{m} \text{skew}_{d_{o,1}} = 0.400$		
		s.d.-a	s.d.-s	cov.	s.d.-a	s.d.-s	cov.	s.d.-a	s.d.-s	cov.
0.4	50	0.062	0.061	0.912	0.088	0.088	0.882	0.098	0.099	0.910
	100	0.044	0.043	0.937	0.063	0.062	0.915	0.070	0.070	0.933
	250	0.028	0.028	0.940	0.040	0.040	0.932	0.044	0.045	0.941
	500	0.020	0.020	0.942	0.028	0.028	0.946	0.031	0.031	0.948
	1000	0.014	0.014	0.951	0.020	0.020	0.946	0.022	0.022	0.948
0.8	50	0.111	0.101	0.857	0.167	0.159	0.783	0.186	0.185	0.863
	100	0.080	0.076	0.898	0.121	0.117	0.850	0.134	0.133	0.902
	250	0.051	0.049	0.933	0.077	0.076	0.908	0.086	0.086	0.930
	500	0.036	0.036	0.940	0.055	0.054	0.930	0.061	0.061	0.943
	1000	0.025	0.025	0.945	0.039	0.039	0.940	0.043	0.043	0.946

C.3 Serial Dependence for Ordinal Time Series

Some simulation results concerning $\hat{\kappa}_{d_{o,1}}(1)$ are shown in Table C.5, again for rank counts with Bin(4, 0.3)-marginal, also see Tables C.1 and C.4. Under the null of serial independence ($\rho = 0$), the asymptotic approximation according to Theorem 7.2.1 apparently works quite well. If using $\hat{\kappa}_{d_{o,1}}(1)$ as a test statistic for the null hypothesis of serial independence (5 % level), where the critical value is computed from Theorem 7.2.1 by plugging-in the sample CDF $\hat{f}$, then the tabulated rejection rates show that the sizes ($\rho = 0$) are close to 0.050, whereas the power values quickly increase with increasing ρ (and certainly with increasing sample size n).

Table C.5: Mean (m.) and standard deviation (s.d.), asymptotic (-a) vs. simulated (-s) of $\hat{\kappa}_{d_{o,1}}(1)$ for i.i.d. Bin(4, 0.3) rank counts, and rejection rates of null hypothesis of serial independence (5 % level) for Bin(4, 0.3)-AR(1) rank counts with autocorrelation parameter ρ .

n	$\hat{\kappa}_{d_{o,1}}(1)$				rejection rate; $\rho =$				
	m.-a	m.-s	s.d.-a	s.d.-s	0.0	0.2	0.4	0.6	0.8
50	-0.020	-0.020	0.096	0.101	0.056	0.254	0.738	0.980	0.999
100	-0.010	-0.010	0.068	0.069	0.052	0.443	0.961	1.000	1.000
250	-0.004	-0.004	0.043	0.044	0.051	0.824	1.000	1.000	1.000
500	-0.002	-0.002	0.030	0.030	0.050	0.981	1.000	1.000	1.000
1000	-0.001	0.000	0.021	0.022	0.052	1.000	1.000	1.000	1.000

D Examples of Distance-based Measures

Table D.1: Overview of some distance-based measures for ordinal random variables using either $d = d_{o,1}$ from (2), or $d = d_{o,2}$ from (3).

	$d = d_{o,1}$	$d = d_{o,2}$
$x_{\text{loc},d}$	median	$s_{\text{round}(E[I])}$
normalized disp_d	$\frac{4}{m} \sum_{k=0}^{m-1} f_k(1 - f_k)$	$\frac{4}{m^2} V[I]$
normalized asym_d	$\frac{1}{m} \sum_{j=0}^{m-1} (1 - f_j - f_{m-j-1})^2$	$(1 - \frac{2}{m} E[I])^2$
normalized skew_d	$\frac{2}{m} \sum_{i=0}^{m-1} f_i - 1 = 1 - \frac{2}{m} E[I]$	
$\kappa_d(X, Y)$	$\frac{2 \sum_{i=0}^{m-1} (f_{XY;ii} - f_{X;i} f_{Y;i})}{\sum_{i=0}^{m-1} (f_{X;i} + f_{Y;i} - 2 f_{X;i} f_{Y;i})}$	$\frac{2 \text{Cov}[I_X, I_Y]}{V[I_X] + V[I_Y] + (E[I_X] - E[I_Y])^2}$
$\kappa_d(h)$	$\frac{\sum_{i=0}^{m-1} (f_{ii}^{(h)} - f_i^2)}{\sum_{i=0}^{m-1} f_i(1 - f_i)}$	$\rho_I(h)$